

INTERN TAKE-HOME ASSIGNMENT
Screening Challenge —AI

Factory Compliance & Alert Escalation System

Design and implement an end-to-end automated compliance system that ingests raw factory video, parses a regulatory policy document, and drives real-time alert workflows.

Dataset	https://www.kaggle.com/datasets/trnhnggiang/video-dataset-for-safe-and-unsafe-behaviours
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1. CHALLENGE OVERVIEW & CORE OBJECTIVE

This challenge asks you to build a complete, automated Factory Compliance and Alert Escalation System. The system must bridge three domains that are rarely integrated in a single pipeline: computer vision (detecting what is happening in a video), natural language understanding (extracting compliance rules from an unstructured regulatory document), and operational workflow automation (routing events to the correct alert or logging channel based on risk severity).

There is no prescribed architecture. You are free to choose your own frameworks, models, and design patterns. The system will be evaluated as a complete functional system.

CORE OBJECTIVE

Ingest raw factory video feeds. Parse a formal EHS compliance policy document. Detect behavioral violations in the video against the policy rules. Classify each violation by severity. Route alerts and logs accordingly. Present all of this through a live operations dashboard.

1.1 Architectural Freedom

You may implement this system using any combination of technologies. Reasonable choices might include (but are in no way limited to):

- Vision models: fine-tuned object detectors, zero-shot vision-language models, or classical computer vision techniques.
- Policy parsing: LLM-based rule extraction, retrieval-augmented generation (RAG), structured prompt engineering, or manual rule transcription backed by a structured schema.
- Backend: Python (FastAPI, Flask, Django), Node.js, or any framework that supports a real-time event pipeline.
- Frontend / Dashboard: React, Vue, Streamlit, Gradio, Dash, or any functional web interface.
- Storage: SQLite, PostgreSQL, MongoDB, or any persistent store appropriate to the data volume.

2. PROVIDED INPUTS

2.1 Raw Video Clips

You will receive a collection of Full HD (1920×1080) video clips captured from fixed-angle security cameras installed in an active manufacturing facility. The clips have the following characteristics:

Property	Value
Resolution	1920 × 1080 (Full HD)
Clip Duration	3 to 18 seconds per clip
Camera Type	Fixed-angle IP security cameras
Environment	Indoor manufacturing facility — active production floor
Content	Ambient shop-floor traffic, forklift movements, human-machine interactions, pedestrian routing, and stationary equipment states

2.2 Compliance Policy Document

You will also receive the facility's Occupational Health and Safety Compliance Policy Manual as a PDF. This is the authoritative source of truth for all compliance decisions your system makes. Key characteristics of this document:

- It defines the facility's behavioral compliance framework across four operational domains: pedestrian movement, equipment intervention, electrical panel management, and forklift load management.
- For each domain, it specifies both the required (safe) behavior and the prohibited (unsafe) behavior, along with the observable indicators that distinguish them.
- It describes the automated monitoring and alerting infrastructure installed at the facility, including the types of alerts triggered upon detection and the reporting format used.

Your system must parse this document to extract the compliance rules that drive your detection and severity classification logic.

3. FUNCTIONAL MODULES

Your system must implement all five of the following modules. Each module has defined inputs, outputs, and interface contracts with the other modules. The modules must function as an integrated pipeline, outputs from one feed into the next.

Module 1 — Detection Engine

ROLE

Ingest video clips and identify behavioral compliance violations by cross-referencing observed activity against the rules extracted from the compliance policy document.

Required Behavior

- Accept raw video clips as input.
- Process each clip frame-by-frame or as a temporal sequence as required by your detection approach.
- Identify and localize any behavioral violations belonging to the compliance categories defined in the policy document.

- For each detected violation, produce a structured detection record containing at minimum: clip identifier, timestamp of occurrence within the clip, the specific compliance rule breached, a description of the observed behavior, and the zone or location within the frame.

Policy Grounding Requirement

The detection logic must be grounded in the compliance policy document. Specifically:

- The behavioral categories your detector operates on must be derived from the policy document through your parsing pipeline, not manually transcribed as hard-coded strings.
- The observable indicators used to classify a behavior as safe or unsafe (e.g., vest color, block count, panel state, walkway position) must be traceable to the relevant policy section.

NOTE

You are not required to achieve perfect detection accuracy on all clip types. However, your detection pipeline must be functional and produce meaningful outputs across the provided clip set. Document your model selection rationale and any known limitations in your repository README.

Module 2 — Severity Categorization Matrix

ROLE

Evaluate the context of each detected violation and assign it a risk severity tier using a classification schema derived from the compliance policy.

Severity Tier Schema

Your system must implement a risk-tier hierarchy with at least the following four levels. The assignment of violations to tiers must be driven by the compliance policy content — specifically, the policy's descriptions of each behavior's hazard context, frequency data, and alerting language:

Tier	Label	General Criteria
LOW	Low Risk	Condition observed but no immediate personnel proximity or imminent hazard. Typically a state-based finding (e.g., equipment condition) with no concurrent personnel exposure.
MED	Medium Risk	Behavioral deviation observed. Personnel present but not in immediate danger. Policy breach confirmed but hazard not yet acute.
HIGH	High Risk	Active unsafe behavior with concurrent personnel exposure or operational risk. Hazard is present and could result in injury.
CRIT	Critical Risk	Immediate danger condition. Most severe policy breach; either high-frequency recurrence, direct injury risk, or the behavior category is explicitly flagged as the highest-consequence hazard in the policy.

Module 3 — Escalation Pipeline

ROLE	Execute programmatic workflows tied to each violation's severity tier, routing events to the correct downstream channel in real time.
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Routing Logic

The escalation pipeline must implement the following mandatory routing rules:

Severity	Required Route	Implementation Requirement
LOW / MEDIUM	Persistent database log	Write a structured violation record to your database (see Module 4 for required fields). No real-time alert required.
HIGH / CRITICAL	Real-time alert trigger AND persistent database log	Trigger an immediate alert in addition to the database log. The alert must be implemented as one of: (a) a simulated visual/audible strobe notification on the dashboard, or (b) a message pushed to a live notification queue (e.g., WebSocket, SSE, or in-process pub/sub).

Module 4 — Automated Report Generation

ROLE	Autonomously produce structured, immutable compliance records for every detected violation, forming the facility's audit trail.
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Required Report Fields

Every violation detected by Module 1 must generate a compliance report. Reports must be written automatically — not manually authored. Each report must contain at minimum the following fields:

Field	Description	Format / Constraint
event_id	Unique identifier for this violation event.	UUID or comparable unique string.
timestamp	Wall-clock time at which the violation was detected.	ISO 8601 (e.g., 2022-11-25T10:45:00Z).
clip_id	Identifier of the source video clip.	Filename or assigned clip reference.
zone	Location or zone identifier within the facility where the event occurred.	String label (e.g., 'Zone-1', 'Zone-2', or inferred from clip metadata).
behavior_class	The specific unsafe behavior detected.	One of the four behavior classes defined in the compliance policy.
policy_rule_ref	Reference to the specific section of the compliance policy that defines the violated rule.	Section identifier string (e.g., 'Section 3.3.2').

Field	Description	Format / Constraint
event_description	Human-readable description of what was observed in the clip.	Free text, minimum 1 sentence.
severity	The severity tier assigned by Module 2.	One of: LOW, MEDIUM, HIGH, CRITICAL.
escalation_action	The routing action taken by Module 3.	String describing the action taken (e.g., 'Logged to DB', 'Real-time alert triggered + DB log').

Output Formats

Reports can be written in one of the following persistent formats:

- JSON: one file per event, or a append-only JSON log file.
- CSV: an append-only audit CSV with the fields above as columns.
- Structured database records: rows in a relational or document store accessible for export.

Module 5 — Operations Dashboard

ROLE

Provide a functional GUI or web application for facility overseers to monitor compliance status, review alerts, and export audit records.

Required Dashboard Features

The dashboard must implement all three of the following functional views:

View A — Live Feed Monitor

- Display a live or simulated feed of processed video clips with compliance status indicators overlaid or adjacent to the clip view.
- Status indicators must reflect the current detection state: no violation detected, violation detected (with severity tier color coding), or alert active.
- For High and Critical events, the dashboard must render the real-time alert triggered by Module 3 (e.g., a visual flash, banner, or modal notification).

View B — Alert Timeline Stream

- Display a real-time or near-real-time chronological stream of compliance events as they are detected.

View C — Historical Log & Export

- Provide access to the full historical compliance log persisted by Module 4.
- Allow the user to filter records by date range, severity tier, and behavior class.
- Provide a functional export button that downloads the filtered or full log in at least one of the report formats specified in Module 4 (JSON, CSV, or database dump).

NOTE

The dashboard does not need to be production-polished in terms of visual design. It must, however, be functional.

4. DELIVERABLES & SUBMISSION REQUIREMENTS

4.1 Repository Structure

Submit a single Git repository (GitHub, GitLab, or equivalent) containing the following:

```
factory-compliance-system/  
├── README.md                # Setup, run instructions, architecture  
├── overview  
│   ├── compliance_policy.pdf # Copy of the provided policy document  
│   ├── data/                # Directory for input video clips  
│   └── src/  
│       ├── detection/       # Module 1 – Detection Engine  
│       ├── severity/        # Module 2 – Severity Categorization Matrix  
│       ├── escalation/      # Module 3 – Escalation Pipeline  
│       ├── reports/         # Module 4 – Automated Report Generation  
│       └── dashboard/       # Module 5 – Operations Dashboard  
├── outputs/                 # Generated reports / logs  
└── requirements.txt / pyproject.toml / package.json # Dependency manifest
```

This directory structure is a recommendation, not a strict requirement. You may organize your repository differently, but the README must make the structure navigable.

5. HINTS & POINTS TO THINK ABOUT

The following are not instructions — they are ideas to help you think through the harder parts of the problem. You may find some useful, others irrelevant depending on your approach.

On Policy Parsing

- The compliance policy defines each unsafe behavior in terms of an observable visual indicator (e.g., a vest color, a block count, a panel state, a spatial boundary). What is the cleanest way to extract these indicator-to-class mappings from prose text?
- The policy's hazard context sections, warning callouts, and frequency data are signals about severity. How does the language differ between a section describing a 'WARNING' condition and one describing a 'CRITICAL SAFETY NOTICE'?
- If you use an LLM to parse the policy, how will you verify that its extracted rules are faithful to the source document? What could go wrong, and how would you catch it?

On Detection

- Some of the four compliance classes are easier to detect from video than others. Consider which are state-based (a condition that persists across many frames) versus action-based (a brief behavioral event). Does this affect your choice of model or approach?
- The policy describes observable indicators for each behavior. How close is your vision model's internal representation to those human-readable indicators? Does it need to be?
- If your detector is uncertain about a borderline case (e.g., a forklift carrying what might be two or three blocks), how should your system handle that? Does the policy give you any guidance?

On Severity and Escalation

- Two of the four behavior categories appear in the compliance policy under a 'WARNING' callout and two appear under a 'CRITICAL SAFETY NOTICE' callout.
- The escalation pipeline's routing rules are binary (Low/Medium → log; High/Critical → alert). What happens if you detect multiple simultaneous violations of different severities in the same clip? How should the pipeline handle that?

On the Dashboard

- The compliance policy document describes the real deployed system's dashboard interface, and what it displays (live camera feed, text log of events). This may inform your dashboard design.
- The alert timeline stream needs to be filterable. Consider how your data model supports filtering efficiently.

6. QUICK REFERENCE SUMMARY

Module	Core Function
1 — Detection Engine	Ingest video; detect behavioral violations against policy-extracted rules.
2 — Severity Matrix	Classify each violation into LOW / MEDIUM / HIGH / CRITICAL tier using policy signals.
3 — Escalation Pipeline	Route LOW/MED to DB log; route HIGH/CRIT to real-time alert + DB log.
4 — Report Generation	Auto-write immutable structured compliance records with all required fields.
5 — Dashboard	Live feed view + alert timeline stream + historical log export.

Submission Checklist

Before submitting, verify each of the following:

	Item
☐	Git repository is accessible to reviewers.
☐	README.md contains setup instructions, architecture description, policy parsing approach, and severity mapping rationale.
☐	(Recommended) Demo video walkthrough recorded and linked.

Good luck. We look forward to seeing what you build.